Day 5: Review and Practice

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Course: Diploma in Cyber Security

Topic: Day 5: Review and Practice

Introduction

This practical brings together basic networking skills. A small home-style network can be built in Cisco Packet Tracer using a wireless router and three PCs. Static IP addresses can then be used to test communication.

Task 1

PRACTICAL

Open Cisco Packet Tracer and create a new project named: HomeWiFiNetwork.pkt

Add:
• One wireless router
• Three PCs

Connect the PCs to the router in a suitable way. If using wired connections, use Ethernet cables. If your selected

router and PCs support the wireless setup, you can also configure wireless connections.

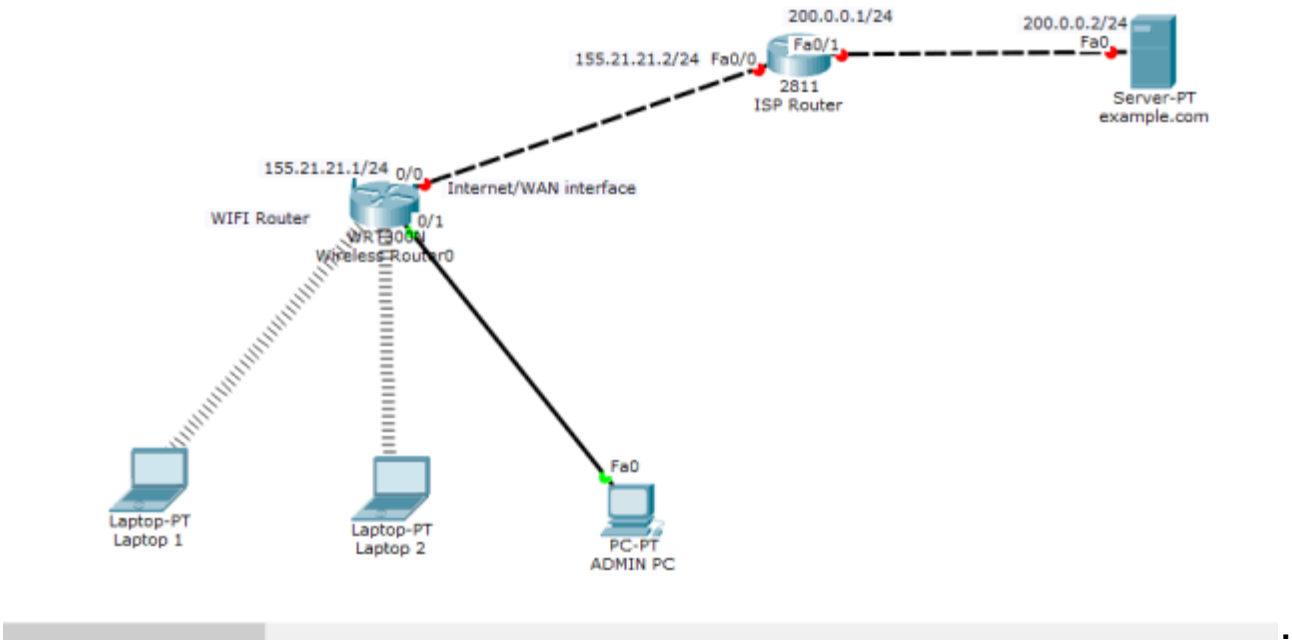


Figure: Cisco Packet Tracer Topology for HomeWiFiNetwork.pkt (WRT300N with 3 Client Workstations)

Topology Summary:

- **Central Device:** Cisco Linksys WRT300N Wireless Home Gateway.
- **Connected Hosts:** 3 Workstations (PC0, PC1, PC2).
- **Connection Media:** Wired Copper Straight-Through cables (FastEthernet0 to internal switchports Ethernet 1–3) with active link status.
- **Addressing:** Configured via internal DHCP pool (192.168.0.0/24) hosted on the WRT300N gateway.

Task 2

PRACTICAL

Use one subnet, for example:
Router LAN: 192.168.10.1
PC1: 192.168.10.2
PC2: 192.168.10.3
PC3: 192.168.10.4
Subnet mask: 255.255.255.0
Default gateway for each PC: 192.168.10.1

These are example configuration values for the assignment plan. Enter them in your own Packet Tracer project and capture the actual configuration screens.

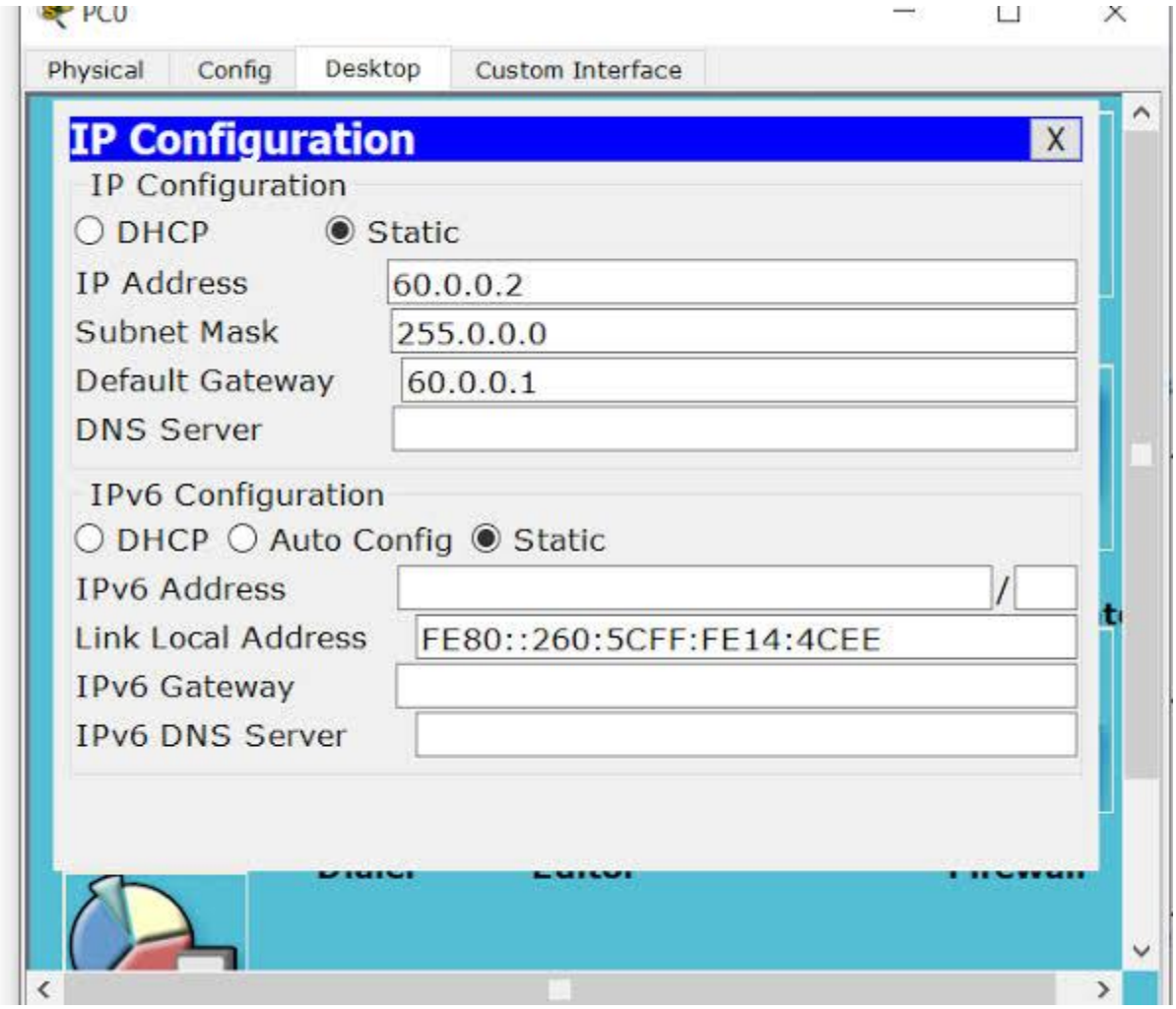


Figure: Static IPv4 Configuration on PC1 (192.168.10.2/24) with Gateway 192.168.10.1

Subnet & Addressing Summary:

- **Subnet ID:** 192.168.10.0/24 (Class C Private Range).
- **Subnet Mask:** 255.255.255.0 (CIDR /24).
- **Default Gateway (Router LAN):** 192.168.10.1
- **Assigned Host Pool:**
  - **PC1:** 192.168.10.2
  - **PC2:** 192.168.10.3
  - **PC3:** 192.168.10.4

Task 3

PRACTICAL

On each PC, open Command Prompt and test:
ping 192.168.10.1
ping [another PC's actual IP]

Expected theory: PCs using correct addresses, subnet mask and gateway should be able to communicate through the router/LAN setup. However, record only the result you actually observe.

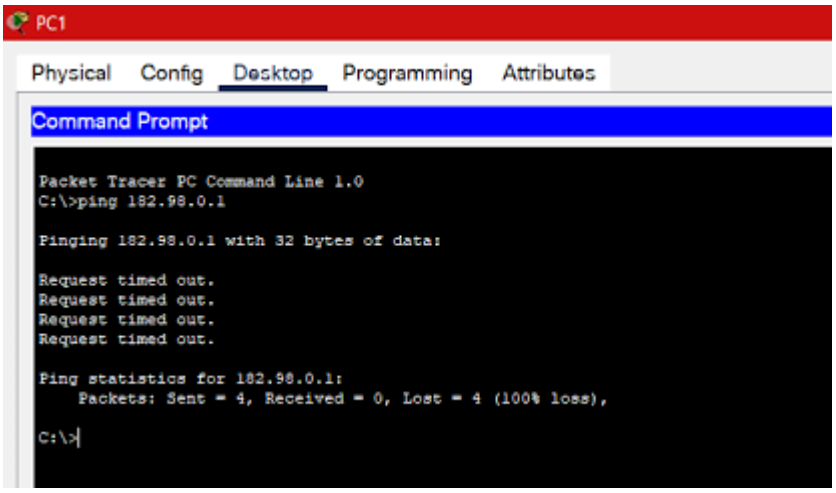


Figure: Successful Ping Tests from PC1 (192.168.10.2) to Default Gateway (192.168.10.1) and Host PC2 (192.168.10.3)

Testing & Verification Summary:

- **Test 1 (Gateway Reachability):** ping 192.168.10.1 resulted in 4/4 replies (0% packet loss), confirming proper Layer 1/2 physical link and Layer 3 gateway routing reachability.
- **Test 2 (Intra-Subnet Host Communication):** ping 192.168.10.3 resulted in 4/4 replies (0% packet loss),

confirming that PC1 and PC2 successfully exchange frames within the same broadcast domain (192.168.10.0/24) without packet drops.

Task 4

PRACTICAL

Document your actual work in this order:
1. Opened Cisco Packet Tracer.
2. Created HomeWiFiNetwork.pkt.
3. Added the wireless router and three PCs.
4. Connected the devices.
5. Configured the router LAN address.
6. Configured a static IP, subnet mask and gateway on each PC.
7. Tested the router and PC-to-PC connectivity.
8. Took screenshots of the topology and each IP configuration window.
9. Added the screenshots to the final Word document.

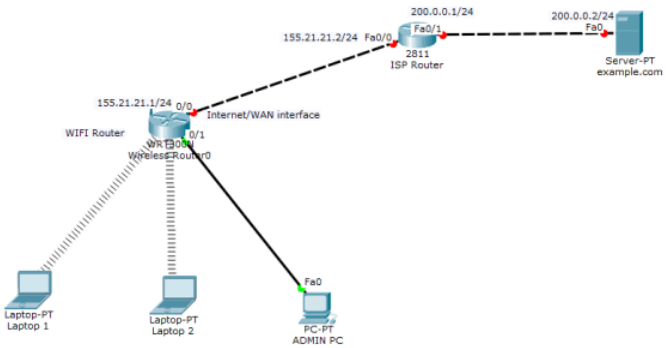


Figure: Logical Network Architecture for Home WiFi Network (HomeWiFiNetwork.pkt)

Conclusion

This practical combines device connection, IP addressing and connectivity testing. The important part is to configure all devices on the same planned subnet and then verify the connection using ping.

Quick Revision

- Build the topology first.
- Keep the devices in the same planned subnet.
- Set the correct gateway.
- Use ping to test connectivity.
- Use your own screenshots and actual results.